

Solvency Field Theory V12.10

The Formation Racetrack:
Inevitable Sorting of Closure Modes
in the Radiation-Dominated Era

A toy formation program from post-shatter
energy ocean to cold-audit survival ecology

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The stage determines the performance. Given a hot dense energy ocean cooling through expansion, any set of closure modes with a mass hierarchy inevitably sorts into the same formation sequence — heavy first, light last, radiation both creating and destroying, and a three-class survival ecology at the cold-audit boundary.

Publication boundary. This document freezes a dimensionless toy formation program. It does *not* predict physical abundances, baryon-to-photon ratios, BBN yields, or CMB spectra. It does not map cadence modes to Standard Model particles. It does not close GR/Boltzmann cosmogenesis. What it does establish: given the V12.8 nine-mode seed and V12.9’s mass hierarchy, a radiation-dominated cooling racetrack inevitably produces ordered formation, disruption-mediated survival windows, and a three-class cold-audit ecology. The inevitability of sorting is the primary result.

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1 The physical picture

Before the shatter, there are no particles. There is energy, moving at c , in an undifferentiated ocean under lag stress. The shatter fragments this ocean. Energy streams in every direction. The universe is hot, dense, and expanding. Everything moves at c because nothing has mass yet.

Mass is not a property energy arrives with. It is what energy *does* when closure traps and focuses it into a stable helical vortex. The nine V12.8 modes are nine lens prescriptions — nine winding patterns that can self-consistently close. V12.9 showed that the focusing strength of each mode determines its mass, with the hierarchy spanning $\sim 339,000\times$ from geometric self-focusing in 3+1 dimensions.

V12.10 asks: given this set of modes and this mass hierarchy, what happens when they encounter a hot cooling environment?

1.1 The racetrack

A radiation-dominated cooling universe is a racetrack. Every closure mode enters the track at the same time (the shatter), but each mode becomes viable at a different temperature — the temperature where the thermal energy density in a Compton-scale volume matches the mode's rest energy. Heavy modes need high energy density. They become viable first, when the universe is hottest. Light modes need low energy density. They become viable last, when the universe has cooled.

The formation sequence is therefore determined by the mass hierarchy. The heaviest mode locks in earliest. The lightest mode locks in latest. The ordering is monotonic with mass and stable across different formation-window models (symmetric, asymmetric, plateau, resonance, turbulent). The mass hierarchy drives the chronology. The formation-window shape does not.

1.2 The radiation double-burden

Radiation plays a dual role. It *creates* vortices by providing the energy density that drives formation — without radiation pressure, the energy ocean would be too diffuse for closure. But it also *destroys* vortices by disrupting nascent patterns before they can complete a full winding cycle.

This double-burden is essential. Removing the radiation drive causes three modes to fail formation entirely ($\rho = 2/3, 3/4, 1$ under asymmetric hot-smash). These are modes that *need* the radiation boost to cross their formation threshold. Without it, their focusing is insufficient.

The disruption side is equally important. Photon-bath disruption is the primary survival filter. Under strong photon disruption, the most fragile mode ($\rho = 3/4$) fails first. Under aggressive full disruption, three modes fail ($2/3, 3/4, 5/4$). The universe creates and destroys simultaneously, and survival depends on which effect wins for each mode.

1.3 Three classes

The integrated ODE freezeout produces a three-class survival ecology at the cold-audit boundary (recombination era):

Robust cold relics ($\rho = 1/2, 4/3, 3/2, 5/3, 2$): five modes that survive across nearly all tested scenarios. They lock in with high relic fractions and persist to the cold boundary. These are the modes that can afford their closure rent comfortably and resist disruption.

Scenario-dependent relics ($\rho = 2/3, 3/4, 5/4$): three modes whose survival depends on the background parameters. Under mild conditions they persist. Under strong disruption they become fragile or fail. Their adequacy margins are tighter and their formation windows are narrower.

Fragile/transient relic ($\rho = 1$): the identity-cadence mode sits at exactly the break-even adequacy threshold ($A_{p5} = 1.0$). It can form but is the most sensitive to perturbation. Under most non-baseline scenarios it becomes fragile.

1.4 Inevitability

The three-class ecology is a generic racetrack feature, not a specific prediction of the exact mode assignments. Null-model controls (R007) showed that shuffling the internal mode properties — which mode gets which mass, which adequacy — produces comparable broad pattern scores. The sorting is inevitable given *any* set of modes with a mass hierarchy on a cooling background.

This is the correct interpretation. The formation racetrack’s claim is not “only these specific modes produce this ecology.” The claim is: “given the V12.8 modes and V12.9 masses, the post-shatter environment inevitably sorts them.” The stage determines the performance. If the stage is right — if the modes are right and the masses are right — then the formation sequence and survival ecology follow automatically. No additional physics is required.

1.5 Abundance is three questions

The toy program discovered that “abundance” is not one question. It splits into at least three:

Number-count: how many quanta of each type? Naturally favors light modes (the lightest mode dominates by count because lighter vortices are cheaper to make).

Energy-capture: how much total energy is trapped in each type? Naturally favors heavy modes (the heaviest mode captures the most energy per vortex).

Relic-occupancy: how much of the available freezeout capacity is used by each type? Distributes broadly across all modes.

Different questions give different answers. The toy correctly refuses to select one as “the” abundance.

2 Formation sandbox and calibration (R001–R003)

R001 built the broad sandbox using V12.8’s nine-mode seed, V12.9’s mass amplitude and closure adequacy, shape/force susceptibility proxies, and a radiation active-gravity factor. All nine modes survived the baseline with lock-in order following the mass hierarchy.

R002 calibrated the background. Sweeping radiation factor, disruption strength, and formation-window width produced survival counts from 5 to 9 modes, with lock-in order 99.8% stable. Removing the radiation double-burden failed $\rho = 2/3$ and $3/4$.

R003 compared five formation-window models. The lock-in order remained 99.1% stable across all variants. The formation chronology is driven by the mass hierarchy, not by the window shape.

3 Disruption and integrated survival (R004–R006)

R004 isolated disruption channels. Photon-bath disruption is the primary survival filter ($\rho = 3/4$ fails first). Hubble tearing reduces margins without changing survival counts. Denominator-weighted and ecology-weighted disruption create partial windows.

R005 replaced instantaneous survival scores with an ODE:

$$\frac{dY_i}{d\tau} = \Gamma_{\text{form},i}(x)(1 - Y_i) - \Gamma_{\text{destroy},i}(x) Y_i, \quad (1)$$

producing integrated relic fractions Y_i for each mode. Heavy modes reach $Y_{\text{peak}} \approx 1.0$; light modes reach $Y_{\text{peak}} \approx 0.6$. The ODE distinguishes robust survivors from fragile/transient modes.

R006 evaluated the ODE at the cold-audit (recombination) marker: five robust, three scenario-dependent, one fragile.

4 Null controls and inevitability (R007–R009)

R007 tested the cold-audit pattern against shuffled and monotonic null controls. The broad pattern score was not null-resistant ($p \approx 0.39$ – 0.98). The exact mode ecology is not yet a specific prediction.

R007B reinterpreted this result: inevitability of sorting is the feature, not the bug. Generic racetrack sorting is expected. The null test confirmed that the sorting does not depend on the specific mode assignments.

R008 confirmed three-class archetype capacity (robust, scenario-dependent, fragile/transient) without mapping modes to particles. R009 rolled up the racetrack program.

5 Abundance proxies and budget conventions (R010–R012)

R010 constructed six budget-convention proxies. Number-count favors $\rho = 1/2$ (84% dominance). Energy-capture favors $\rho = 2$ (91% dominance). Relic-occupancy distributes broadly (effective modes ≈ 8.7). The budget convention determines the abundance profile.

R011 tested abundance proxies against class-preserving nulls. Budget-convention dominance was confirmed. Exact fraction profiles were not claimed.

R012 froze the full program with explicit claim boundaries.

6 What is earned, conditional, and open

6.1 Earned

The formation racetrack inevitably sorts closure modes by mass threshold: heavy first, light last.

The lock-in order is stable across five formation-window models (99% mean order score).

Radiation plays an essential double-burden role: both creating and destroying vortices.

Photon-bath disruption is the primary survival filter.

ODE integration distinguishes robust, scenario-dependent, and fragile/transient relic classes.

Abundance splits into number-count, energy-capture, and relic-occupancy questions with different answers.

Inevitability of sorting is the primary result — any set of modes with a mass hierarchy on this background sorts the same way.

6.2 Not earned

Physical abundances, baryon-to-photon ratio, BBN yields, or CMB spectrum predictions.

Mode-to-particle mapping.

Exact survival ecology as a specific prediction (null-model controls show it is generic).

GR/Boltzmann cosmogenesis closure.

6.3 Open frontier

Physical background replacement: $H(T)$, $a(T)$, thermal thresholds.

Mechanically motivated disruption terms.

Decay channels: how heavy/fragile modes convert to lighter ones.

Connection to observed particle stability and lifetimes.

BBN yield predictions from mode-specific nuclear physics.

CMB spectrum from the un-trapped photon population.

7 Conclusion

V12.10 establishes that the post-shatter formation story is inevitable. Given a set of self-consistent closure modes with a mass hierarchy derived from geometric focusing, and given a radiation-dominated cooling environment, the formation racetrack produces an ordered lock-in sequence, a radiation-mediated survival filter, and a three-class cold-audit ecology — automatically, with no additional physics required.

The inevitability is the result. The stage determines the performance. If the nine modes are the right modes (V12.8) and their masses are the right masses (V12.9), then the formation chronology and survival ecology follow from the thermal history alone. The universe does not choose which particles form first or which ones survive. The geometry chooses. The cooling executes. The outcome is the only outcome the stage permits.

The shatter creates the energy ocean. The cooling creates the racetrack. The mass hierarchy determines the order. The closure adequacy determines who survives. The photons that didn't find a stable vortex are still traveling in straight lines at c , thirteen billion years later. We call them the CMB.

Acknowledgments

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References

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A Gate rollup: R001–R012

Gate	Title	Contribution
R001	Formation sandbox	Broad toy. All 9 modes survive baseline. Lock-in order follows mass hierarchy.
R002	Background calibration	Radiation factor informative. Survival varies 5–9 modes. Order 99.8% stable.
R003	Formation drive refinement	Five window shapes tested. Order 99.1% stable. Mass hierarchy drives chronology.
R004	Disruption localization	Photon bath is primary filter. $\rho=3/4$ most fragile. Hubble tearing reduces margins.
R005	Survival ODE	Integrated freezeout. Heavy modes $Y \approx 1.0$, light modes $Y \approx 0.6$. Robust/fragile split.
R006	Recombination audit	Five robust, three scenario-dependent, one fragile at cold-audit marker.
R007	Null controls	Broad ecology not null-resistant. Exact mode split not special.
R007B	Inevitability interpretation	Generic sorting is the feature. Racetrack sorting preserved as primary result.
R008	Archetype sanity	Three-class capacity confirmed. No particle mapping.
R009	Racetrack rollup	Formation program frozen at racetrack milestone.
R010	Abundance proxies	Number/energy/occupancy give different answers. Budget convention dominates.
R011	Class-preserving nulls	Budget effects localized. Exact fractions open.
R012	Full rollup	Toy program frozen. Physical cosmogenesis open.